

Noncontraction Transport Through Unequal-Length Merges

Collatz proof project — September 5, 2026

Abstract

An unequal-length merging predecessor inherits full noncontraction when its finite prefix is noncontracting and its final coefficient dominates the original prefix coefficient. We formalize this transport principle and an infinite arithmetic family meeting after one versus twelve steps. A bounded exact search supplies additional examples but leaves a global existence obligation. Neither nondivergence nor Collatz is proved.

Let $T(n) = n/2$ for even n and $(3n + 1)/2$ for odd n . Write $j_t(n)$ for the odd count and $C_t(n) = 3^{j_t(n)}/2^t$. Call n noncontracting if $C_t(n) \geq 1$ for every $t \geq 0$.

Theorem 1 (Dominating merge). *Suppose n is noncontracting, $T^s(x) = T^t(n)$, $C_i(x) \geq 1$ for $0 \leq i \leq s$, and $C_s(x) \geq C_t(n)$. Then x is noncontracting.*

Proof. For $v \geq 0$, the coefficient cocycle and the shared endpoint give

$$C_{s+v}(x) = C_s(x)C_v(T^s(x)) = \frac{C_s(x)}{C_t(n)}C_{t+v}(n) \geq 1.$$

Earlier times are covered by the prefix hypothesis. The Lean proof uses natural-number cross multiplication and cancellation of a positive power of three. It does not assume the shifted seed $T^t(n)$ is noncontracting; that property does not follow just by shifting the original hypothesis. $\square$

Theorem 2 (An infinite merging family). *For every integer $Q \geq 0$, put*

$$N = 111 + 4374Q, \quad M = 103 + 4096Q.$$

Then $0 < M < N$, and

$$T(N) = T^{12}(M) = 167 + 6561Q, \quad j_1(N) = 1, \quad j_{12}(M) = 8.$$

Every prefix of the twelve-step path from M is noncontracting. Consequently, if N is noncontracting, so is the smaller seed M .

Proof. Lean's kernel verifies $T^{12}(103) = 167$, $j_{12}(103) = 8$ and all thirteen prefix coefficient inequalities. Binary-cylinder transport preserves these parities for $103 + 2^{12}Q$ and adds 3^8Q to its endpoint. The seed N is odd, giving the same endpoint after one step. The coefficient comparison is

$$\frac{C_{12}(M)}{C_1(N)} = \frac{3^8/2^{12}}{3/2} = \frac{2187}{2048} > 1.$$

The first theorem applies, and $N - M = 8 + 278Q > 0$. $\square$

Thus no least positive noncontracting seed can belong to this progression. All N are multiples of three, and the meeting occurs after an increasing forward step. The result is not a construction of a noncontracting seed, nor an exclusion of noncontraction for every member without the minimality condition. No mathematical priority claim is made.

Finite search. The exact Python search takes the 7,495 seeds $0 < n < 2^{18}$ whose first 18 prefix coefficients are at least one. Equal-time/equal-count replacement already covers 1,391 of them as ordinary smaller merges. Searching both forward and replacement lengths from zero through 18 covers another 3,768. All 3,768 additions have certificates satisfying the first theorem; 937 of the selected certificates use a positive forward time. Each is independently replayed, and no original path descends before its meeting. The 2,336 unresolved seeds begin 27, 327, 447, 495, 639, 667, 703, 763. These are finite seed counts, not coverage of all their cylinder lifts. Failure within these bounds says nothing about longer replacements.

Verification and remaining gap. The three public declarations in `Collatz.DominatingMerge` are included in the axiom audit. The reproducible search is `analysis/variable_length_surgery.py`; its tests compare against every smaller seed at depth eight and replay five members of the family. The census is not kernel certified. A proof must still force a suitable smaller predecessor for every hypothetical least noncontracting seed; the present certificates cover only particular cases. Nontrivial cycles also remain outside the conclusion.

Kernel-certified arithmetic progressions

The individual merging examples can be turned into rules that apply to arbitrarily large seeds. This extension certifies their validity, not universal applicability.

Theorem 3 (Progression lifting). *Let n, x, t, s, a, b be natural numbers satisfying $0 < x < n$, $T^s(x) = T^t(n)$, $C_i(x) \geq 1$ for $0 \leq i \leq s$, and $C_s(x) \geq C_t(n)$. Write $j = j_t(n)$ and $k = j_s(x)$, and assume*

$$3^j a = 3^k b, \quad 2^s b \leq 2^t a.$$

For every $Q \geq 0$, put $N_Q = n + 2^t a Q$ and $X_Q = x + 2^s b Q$. Then $0 < X_Q < N_Q$. If N_Q is noncontracting, so is X_Q .

Proof. Binary-cylinder transport gives

$$T^t(N_Q) = T^t(n) + 3^j a Q, \quad T^s(X_Q) = T^s(x) + 3^k b Q.$$

The balance condition makes these endpoints equal. Prefix parities and odd counts through t and s are unchanged, so the noncontracting-prefix and coefficient-domination checks remain valid. The first theorem therefore transfers full noncontraction. Finally, $x < n$ and $2^s b \leq 2^t a$ give strict ordering for every nonnegative quotient, while $x > 0$ gives positivity. $\square$

Choosing the steps. A reduced choice is

$$a = 3^{\max(k-j, 0)}, \quad b = 3^{\max(j-k, 0)}.$$

It satisfies the balance equation. Coefficient domination gives the required ordering of the two seed increments. The generator shifts a certificate backward along this progression only while both seeds stay positive and the replacement remains strictly smaller. The shifted certificate is checked again, including every replacement-prefix parity. These normalization operations are not assumed correct by Lean: the resulting finite data must independently satisfy its validator.

What the kernel checks. `MergeRule.Valid` records positive seed ordering, the exact common endpoint, every replacement-prefix coefficient inequality, coefficient domination, ternary balance, and ordered seed increments. `MergeRule.sound` proves the theorem above for any valid rule. The generated `MergeRuleTable.rules` contains 360 rules. Its `rules_valid` theorem checks them with ordinary kernel reduction using `decide`; no external verifier result or added axiom supplies validity. `certified_rule_sound` applies the generic theorem to any listed rule and any natural quotient.

Scope of the finite generation. The 3,768 successful dominating certificates from the preceding search normalize to 360 distinct progressions. The inclusion filter removes no further distinct progression in this run. The listed rules include the earlier $111 + 4374Q$ family and simple inverse families such as $2 + 3Q$ merging with $1 + 2Q$. The finite Python search and its completeness remain separate from the kernel-certified validity of the resulting list.

Every listed progression is excluded for a least positive noncontracting seed: its certified replacement would be smaller and noncontracting. No theorem says that this finite union contains every such hypothetical seed. In particular, the earlier unresolved seeds do not acquire an in-budget certificate from this normalization. The missing existence argument may need a seed-dependent, unbounded family of rules. The nontrivial-cycle problem is also unchanged.

Reproduction. Run `python3 analysis/merge_progressions.py --write-lean` from the root, then `lake build Collatz.MergeRuleTable` from `lean/`. The generator's JSON and the Lean source are both stored in the repository. Tests check normalization, progression inclusion with lower bounds, and exact agreement between the generated source and the saved rule data. The theorem audit covers the generic soundness theorem and both public table theorems. No mathematical priority claim is made.